

Proposals, rejections, and Common Ground management[★]

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Abstract. This paper connects two seemingly unrelated phenomena discussed in the literature and offers a unified account: (i) Why must modal adverbs (e.g., English *probably*) be positive and behave like Positive Polarity Items (PPIs) (see Bellert 1977, Nilsen 2004, Piñon 2009, Wolf 2015)? (ii) Why do corrective constructions (e.g., Chinese *bìng bù*, English *not p but q*) obligatorily involve propositional negation (see Toosarvandani 2014, Yuan and Liu 2025, Wu 2025, Zhang 2025)? I argue that **modal adverbs** and **corrective constructions** are speaker-oriented, Common Ground (CG)-management operators, signaling two speech acts distinct from assertions: **proposals** and **rejections**. (i) Based on a proposition p 's being sufficiently likely in speaker S 's epistemic worlds, S uses a modal adverb to **propose** p , supporting (i.e., expressing a bias toward), but not requiring, its being considered for inclusion in the CG. (ii) Speaker S uses a corrective construction to **reject** (i.e., **negate**) a proposition p , thereby requiring its exclusion from the CG. The speaker's support (bias) and requirement can be modeled as existential and universal modality.

Keywords: Speech acts · Common ground · Modality · Modal adverbs · Corrective constructions

1 Introduction

Negation is a universal feature of human language. Intriguingly, some expressions are subject to polarity restrictions, and this becomes particularly striking when the same patterns are found across languages. This paper investigates two such phenomena: (i) **modal adverbs** (e.g., *probably*, see (1)), which must be positive and behave like Positive Polarity Items (PPIs), and (ii) **corrective constructions** (e.g., *not p but q*, see (2)), which require sentential negation.

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- (1) a. Probably, it will rain.
b. #Improbably, it will rain.
- (2) a. Space travel is not possible, but a dream.
b. #Space travel is impossible, but a dream.

Although both phenomena have been studied in the literature, they have largely been investigated independently. As a result, it remains unclear whether their polarity requirements arise from construction-specific properties or reflect a common underlying mechanism. This paper pursues the latter line of inquiry. Specifically, I argue that the polarity restrictions of modal adverbs and corrective constructions arise from, and thus signal, two distinct speech acts.

The study of speech acts, including the relation between them and their linguistic clues, begins with Austin’s (1962) pioneering work. Austin argues that not all declarative sentences perform the same speech act. He identifies **performatives** as utterances that do not describe anything and are therefore neither true nor false. Rather, they perform an action. He also observes that performatives are subject to characteristic linguistic constraints: as illustrated by (3), the verb has to be in the first-person singular present indicative active.

- (3) I name this ship the Queen Elizabeth. **Performative**
 $\rightsquigarrow$ Verb: first person singular present indicative active

In light of Austin’s speech-act theory, I argue that, as paraphrased in (4), a modal adverb like *probably* signals a **proposal**, whereas a corrective construction signals a **rejection**. Proposals and rejections are considered speaker-oriented speech acts that give rise to conversational moves distinct from those of assertions in managing the Common Ground (CG).

- (4) a. ‘Probably p ’ = ‘I propose p ’, or ‘Let’s consider p ’ **Proposal**
b. ‘not p (but q)’ = ‘I reject p ’, or ‘Don’t believe p ’ **Rejection**

Under this view, a speaker S uses a modal adverb to propose a proposition p . Such a proposal is licensed by p ’s being sufficiently likely in S ’s epistemic worlds, but S only supports (i.e., shows a bias toward), rather than requires, the consideration of p for inclusion in the CG. By contrast, a speaker uses a corrective construction to reject the inclusion of p in the CG and requires interlocutors to reject it. The different polarity restrictions of modal adverbs and corrective constructions follow from how proposals and rejections work in CG management.

The current analysis also makes a distinction between the strength of proposals and rejections. Proposals involve **existential modality**, reflecting the speaker’s support or bias without requiring interlocutors to accept the proposed proposition. Rejections, by contrast, involve **universal modality**, reflecting the speaker’s requirement that interlocutors accept the rejection.

Below, §2 presents cross-linguistic data, demonstrating the polarity restrictions of modal adverbs and corrective constructions. §3 develops the analysis. §4 compares the current analysis with existing approaches and discusses theoretical implications and future directions. §5 concludes.

2 Empirical phenomena

This section presents data from English and Chinese to show the polarity restrictions of modal adverbs (§2.1) and corrective constructions (§2.2).

2.1 Modal adverbs

As noted in the literature (Bellert 1977, Nilsen 2004, Piñon 2009, Ernst 2009, Wolf 2015), modal adverbs must be positive and behave like PPIs. Below I use data from both English and Chinese to illustrate their behavior.

First, **modal adjectives** (e.g., *probable*, *evident*) and their negated counterparts can be used symmetrically (see (5)–(6)). By contrast, **modal adverbs** (e.g., *probably*, *evidently*) exhibit an asymmetry: only the positive forms, but not their negated counterparts, can be used felicitously (see (7)–(8)).

- (5) a. It is probable that she will win.
b. It is improbable that she will win.
- (6) a. It is evident that she will win.
b. It is not evident that she will win.
- (7) a. Probably, she will win.
b. #Improbably, she will win.
- (8) a. Evidently, she will win.
b. #Not evidently, she will win.

Chinese exhibits exactly the same pattern (see (9) and (10)). In particular, Chinese does not make a morphological distinction between modal adjectives and adverbs. Instead, the distinction between the two categories is manifested syntactically, e.g., by their position in a sentence. Thus the Chinese data further show that the non-existence of negated modal adverbs cannot be attributed to word-level blocking, but must instead arise from a sentence-level constraint.¹

- (9) a. tā huì yíng de jì-xiàng hěn míngxiǎn
3SG will win LINKER sign very evident
'It is evident that she will win.' ✓evident
b. tā huì yíng de jì-xiàng bù míngxiǎn
3SG will win LINKER sign NEG evident
'It is not evident that she will win.' ✓not evident
- (10) a. hěn míngxiǎn tā huì yíng
very evident 3SG will win
'Evidently, she will win.' ✓evidently
b. * bù míngxiǎn tā huì yíng
NEG evident 3SG will win
Intended: #'Not evidently, she will win.' #not evidently

¹ In Chinese, the positive use of gradable adjectives or adverbs typically requires the presence of a semantically bleached degree modifier *hěn* 'very' (see (9a) and (10a)).

Second, cross-linguistically, modal adverbs cannot scope under negation:

- (11) a. She probably will not win. $\checkmark$ **probably** > **neg**
 b. #She will not probably win. $\#$ **neg** > **probably**
- (12) a. She evidently will not win. $\checkmark$ **evidently** > **neg**
 b. ??She will not evidently win. $\#$ **neg** > **evidently**
- (13) a. tā míngxiǎn bú huì yíng
 3SG evidently NEG will win
 ‘She evidently will not win.’ $\checkmark$ **evidently** > **neg**
 b. * tā bù míngxiǎn huì yíng
 3SG NEG evidently will win
 Intended: #‘She will not evidently win.’ $\#$ **neg** > **evidently**

Third, in English, unlike modal adjectives, modal adverbs sound degraded in yes/no questions:

- (14) a. Is it probable that she will win? $\checkmark$ **probable**+y/n q
 b. ??Will she probably win? $\#$ **probably**+y/n q
- (15) a. Is it evident that she will win? $\checkmark$ **evident**+y/n q
 b. ??Will she evidently win? $\#$ **evidently**+y/n q

Chinese shows a similar pattern. Modal adverbs like *míngmíng* (‘evidently’) are incompatible with regular yes/no questions (see (16b)) and low-negation yes/no questions (see (17b)), but are compatible with high-negation yes/no questions (see (18b)). The position of *míngmíng* differs between (17b) and (18b).

- (16) a. [tā huì yíng] ma?
 3SG will win Q
 ‘Will he win?’ $\checkmark$ y/n q
 b. * [tā míngmíng huì yíng] ma?
 3SG evidently will win Q
 #‘Will he evidently win?’ $\#$ **míngmíng**+y/n q
- (17) a. [tā bú huì yíng] ma?
 3SG NEG will win Q
 ‘Won’t he win?’ $\checkmark$ **low-neg** y/n q
 b. * [tā míngmíng bú huì yíng] ma?
 3SG clearly NEG will win Q
 # ‘Won’t he evidently win?’ $\#$ **míngmíng**+**low-neg** y/n q
- (18) a. tā bú (shì) [huì yíng] ma?
 3SG NEG COP will win Q
 ‘He will win, won’t he?’ $\checkmark$ **high-neg** y/n q
 b. tā bú (shì) [míngmíng huì yíng] ma?
 3SG NEG COP clearly will win Q
 ‘He evidently will win, won’t he?’ $\checkmark$ **míngmíng**+**high-neg** y/n q

The distinction between low- and high-negation yes/no questions is reflected in two diagnostics. (i) the copula *shì* may optionally appear only in high-negation questions (see (17) vs. (18)). (ii) low-negation yes/no questions have rising intonation, whereas high-negation yes/no questions have falling intonation.

The scope behavior of modal adverbs and their incompatibility with regular yes/no questions further suggest that modal adverbs cannot be part of the propositional-level semantics. Instead, modal adverbs operate above the propositional level and are related to the speaker’s subjective attitude toward the proposition (see Ye 2026 on high-negation yes/no questions).

To sum up, in natural language, modal adverbs resist being negated, and their meaning contribution must lie above the level of propositional semantics.

2.2 Corrective constructions

Cross-linguistically, corrective constructions include English *but* (see Toosarvandani 2014) and Chinese *bǐng bù* (see Yuan and Liu 2025, Wu 2025, Zhang 2025). Intuitively, corrective constructions are used to counter, or correct, an expectation. They exhibit a polarity restriction that is exactly opposite to that of modal adverbs: they obligatorily require sentential negation.

As illustrated by (19), English *but* has at least four different uses. According to Toosarvandani (2014), the first three uses in (19) share a unified semantics of *but*, whereas the exceptive use (see (19d)) is distinct. The examples in (19) also show that, among these four uses, only the corrective use in (19b) necessarily involves negation; the other three do not.

- | | | | |
|------|----|--|---|
| (19) | a. | The player is tall, <u>but</u> agile. | Counterexpectational <i>but</i> |
| | b. | Liz doesn’t dance, <u>but</u> sings. | Corrective <i>but</i>: not <i>p</i> but <i>q</i> |
| | c. | John is tall, <u>but</u> Bill is short. | Semantic opposition |
| | d. | Everyone <u>but</u> Marion has fed Dudley. | Exceptive <i>but</i> |

In the corrective construction *not p but q*, the presence of negation is obligatory. Therefore, when *p* already contains negation, the double negation part ‘not *p*’ cannot be reduced to a positive sentence, as illustrated by (20).

- | | | |
|------|----|---|
| (20) | a. | It is <u>not</u> that he didn’t come, <u>but</u> that he came and left immediately. |
| | b. | #It is that he came, <u>but</u> that he came and left immediately. |

Furthermore, the required negation in *not p but q* must be sentential (propositional), not morphological. As illustrated by (21) and (22), *not like* and *not possible* cannot be replaced by their morphologically negated counterparts, *dislike* and *impossible*.

- | | | |
|------|----|--|
| (21) | a. | Max doesn’t like spinach <u>but</u> chard. |
| | b. | #Max <u>dislikes</u> spinach <u>but</u> chard. |
| (22) | | Toosarvandani (2014): (69) |
| | a. | Space travel is <u>not</u> possible, <u>but</u> a dream. |
| | b. | #Space travel is <u>impossible</u> , <u>but</u> a dream. |

The four uses of English *but* (see (19)) correspond to different lexical items in Chinese, as illustrated in (23). Specifically, *bìng bù* corresponds to the negation component of English corrective construction *not p but q* (see (23b)).

- (23) a. zhè wèi xuǎn-shǒu hěn gāo, dàn hěn líng huó
this CL player very tall but very agile
'The player is tall, but agile.' **dàn: counterexpectational use**
- b. Liz bìng bù tiào-wǔ (dàn tā chàng gē)
Liz BÌNG NEG dance but 3SG sing song
'Liz doesn't dance, (but sings).' **bìng bù: corrective use**
- c. John hěn gāo, ér Bill hěn āi
John very tall but/and Bill very short
'John is tall, but Bill is short.' **ér: semantic opposition**
- d. chú-le Marion měi gè rén dōu wèi-le Dudley
except Marion every CL person all feed-PRF Dudley
'Everyone but Marion has fed Dudley.' **chú-le: exceptive use**

Chinese corrective construction *bìng bù* consists of (i) *bìng*, a conjunction particle similar to English *and* (see (24)), and (ii) *bù*, a negation particle that can be replaced by other negation particles (e.g., *méi*, *fēi*).

- (24) tā kàn-le yī běn shū, bìng zuò-le bǐ-jì
3SG read-PRF one CL book AND make-PRF note(s)
'She read a book and took notes.'

Like the English counterexpectational use of *but*, Chinese *dàn* does not require negation (see (25)). By contrast, like English corrective construction *not p but q*, Chinese *bìng bù* obligatorily requires negation. Therefore, when the clause following *bìng bù* already contains negation, the use of *bìng bù* results in double negation, which cannot be reduced to a positive sentence (see (27) vs. (28)).

- (25) wài-miàn zài xià-yǔ, dàn wǒ (méi) yǒu sǎn
outside at rain but 1SG NEG have umbrella
'It's raining outside, but I have no/an umbrella.' ✓ **dàn (+neg)**
- (26) wài-miàn zài xià-yǔ, dàn wǒ bìng méi yǒu sǎn
outside at rain but 1SG BÌNG NEG have umbrella
'It's raining outside, but I have no umbrella.' ✓ **bìng+neg**
- (27) * wài-miàn zài xià-yǔ, dàn wǒ bìng yǒu sǎn
outside at rain but 1SG BÌNG have umbrella
Intended: 'It's raining outside, but I have an umbrella.' # **bìng neg**
- (28) wài-miàn zài xià-yǔ, dàn wǒ bìng bú shì méi yǒu sǎn
outside at rain but 1SG BÌNG NEG COPULA NEG have umbrella
'..., but it's not that I have no umbrella.' ✓ **bìng+double neg**

Like English corrective construction *not p but q*, the required negation in Chinese *bìng bù* must be sentential (propositional), not morphological. As illustrated in (29) and (30), *not necessary* cannot be replaced by *unnecessary*.

- (29) jīn-tiān bìng bù/fēi yīdìng huì xià yǔ
 today BÌNG NEG necessary will fall rain
 ‘It’s not necessary that it will rain today.’ ✓ *bìng*+sentential neg
- (30) * jīn-tiān bìng wèi-bì huì xià yǔ
 today BÌNG unnecessary will fall rain
 Intended meaning is the same as (29) #*bìng*+morphological neg

To sum up, in natural language, the use of corrective constructions obligatorily requires sentential negation.

3 Analysis

This section begins with a brief review of previous work on modal adverbs, which suggests that a successful analysis must take into account their speaker-orientedness (§3.1). In light of this, I analyze modal adverbs and corrective particles as speaker-oriented, CG-management operators that realize two distinct speech acts, as paraphrased in (4): proposals and rejections (§3.2 and §3.3). §3.4 highlights the distinction among three speech acts: assertions vs. proposals vs. rejections. §3.5 further examines the interaction between proposals and rejections (e.g., the rejection of what is proposed).

3.1 A brief review of previous work on modal adverbs

The study of modal adverbs in formal linguistics dates back to Bellert (1977). According to Bellert (1977), as paraphrased in (31), modal adverbs are ‘predicates whose argument is the truth of the proposition expressed by the respective sentence (not the fact, event, or state of affairs denoted by the sentence in question)’ (p. 343), whereas corresponding modal adjectives are ‘predicates over the fact, event, or state of affairs referred to by the sentence’ (p. 345).

Under this view, only the truth of a proposition p is qualified for degree evaluation. Thus *probably p* means a high probability that p is true (see (32)). However, it is unclear why *improbably p* cannot be used to express a low probability that p is true. Thus, the unacceptability of *improbably p* is not explained.

- (31) **Bellert (1977)’s analysis of modal adverbs vs. modal adjectives:**
 a. *Probably p* = It is probably true that p .
 b. *It is (im)probable that p* = The fact described by p is (im)probable
- (32) (31a) $\rightsquigarrow$ the truth of p has a high probability

Nilsen (2004) argues that modal adverbs and their corresponding modal adjectives are not synonymous but differ in semantic strength. This claim is based on the contrast in (33): (33a) is intuitively felicitous, whereas (33b) sounds contradictory. Specifically, Nilsen (2004) argues that *possibly* is semantically stronger than *possible*. Consequently, the weaker statement *It is possible that p* can be followed by the stronger statement *certainly* $\neg p$ without contradiction, whereas the two strong statements *possibly p* and *certainly* $\neg p$ are incompatible.

- (33) a. It's possible that Le Pen will win, even though he certainly won't.
 b. #Le Pen will possibly win, even though he certainly won't.

However, it remains unclear why modal adverbs are semantically stronger than their corresponding modal adjectives (see also Piñon 2009, Wolf 2015).

Piñon (2006), Ernst (2009) and Wolf (2015) argue that modal adverbs are speaker-oriented. As paraphrased in (34), Ernst (2009) claims that modal adverbs represent 'speakers' asserting an evaluation of a proposition, about the degree of certainty about that proposition's truth' (p. 540). However, this analysis faces the same problem as Bellert (1977): it fails to explain why *improbably* *p* is unattested and does not indicate that the speaker considers *p* improbable.

- (34) *Probably* *p* = the speaker takes *p*'s truth as probable (Ernst 2009)

As paraphrased in (35), Piñon (2006) claims that modal adverbs modify the sincerity condition of assertions. Under this analysis, the absence of *improbably* *p* follows naturally, since its use would violate the sincerity condition.

- (35) *Probably* *p* = the speaker asserts *p* with a sincerity condition that the degree of his belief in the truth of *p* reaches a norm, *d* (Piñon 2006)

In light of this, Wolf (2015) further distinguishes between modal adverbs and modal adjectives: modal adverbs express a speaker-oriented, subjective estimate of probability, whereas modal adjectives express objective probability (see (36)). As Wolf (2015) puts it, 'Possibility modal adverbs are used as hedges that allow the speaker to assert a proposition with a lesser force, thereby insuring the speaker from losing credibility in case of a falsehood' (p. 140).

- (36) a. It's probable that the dog is on the lawn.
 the speaker asserts the proposition *it's probable that the dog is on the lawn* with a degree of strength = 1 $\rightsquigarrow$ **objective probability**
 b. The dog is probably on the lawn.
 the speaker asserts the proposition *the dog is on the lawn* with a degree of strength > 0.5 $\rightsquigarrow$ **subjective estimate of probability**

Thus, for both Piñon (2006) and Wolf (2015), modal adverbs indicate a speaker's degree of commitment to what they say, and this commitment must be sufficiently high, which licenses positive modal adverbs (e.g., *probably*) and rules out negative ones (e.g., *improbably*).

However, such a view violates the Gricean maxims. According to the Maxims of Quality and Manner (see (37)), an assertion with less force indicates less-than-full commitment and therefore should not occur. In other words, from a pragmatic perspective, if a speaker is not 100% committed to what they say, they had better remain silent.

- (37) a. **Maxim of Quality:** Do not say what you believe is false. Do not say that for which you lack adequate evidence.
 b. **Maxim of Manner:** Be brief.

The dilemma between seemingly less-than-full commitment (as indicated by the use of a modal adverb) and the very act of assertion cannot be resolved within the speech act of assertion. We need to distinguish a speaker’s commitment to what they say from what they expect their utterance to accomplish. Accordingly, I analyze *probably p* as a distinct speech act, **proposal**, rather than an assertion.

3.2 Proposal as a speech act

The current analysis combines the insights that (i) the use of modal adverbs is speaker-oriented (Piñon 2006, Ernst 2009, Wolf 2015) and (ii) different speech acts are associated with different conversational moves for managing the Common Ground (CG) (see, e.g., Stalnaker 1978, Romero and Han 2004, Farkas and Bruce 2010, Frana and Rawlins 2019).

The intuition is that modal adverbs make an utterance more confined to the speaker and therefore less direct in its effect on the CG than an assertion. Although the speaker remains fully committed to what they say, the conversational move performed by a proposal does not require interlocutors to respond to the proposition in the way that an assertion does.

To capture this difference between the conversational moves associated with proposals and assertions, I distinguish three statuses of a proposition (see (38)).

- (38) Three statuses of a proposition *p*:
- a. **Publicly displayed:** *p* becomes publicly visible and supported by the speaker, but is not yet a direct candidate for response
 - b. **Publicly advanced:** *p* becomes a direct candidate for interlocutors’ response (e.g., confirming, reversing)
 - c. **Publicly accepted:** *p* is incorporated into the CG

An **assertion** publicly advances a proposition for interlocutors’ response, naturally licensing self-confirmation but resisting self-doubt, whereas a **proposal** merely displays a proposition while expressing the speaker’s support for it, licensing self-doubt but resisting self-confirmation (see (39) vs. (40)). Moreover, while both asserted and proposed propositions can be refuted, full acceptance from an interlocutor is natural only in the case of assertions (see (41) and (42)).

- | | | |
|------|---|--------------------------------------|
| (39) | a. ??It rained. Who knows? | Assertion: #self-doubt |
| | b. It probably rained. Who knows? | Proposal: ✓self-doubt |
| (40) | a. It rained. I know it did. | Assertion: ✓self-confirmation |
| | b. ??It probably rained. I know it did. | Proposal: #self-confirmation |
| (41) | A: Mary came. | Mary came is asserted. |
| | a. B: You are wrong. She didn’t. | ✓rebuttal |
| | b. B: I’ll take your word for it. | ✓full acceptance |
| (42) | A: Probably Mary came. | Mary came is proposed. |
| | a. B: You are wrong. She didn’t. | ✓rebuttal |
| | b. #B: I’ll take your word for it. | #full acceptance |

Based on the above observations, (43) informally characterizes the semantics of a proposal (*probably p*) and contrasts it with that of a regular assertion (*p*).

(43) **The semantics of a proposal, *probably p*:**

Publicly advanced (asserted)		Publicly displayed (presupposed)
<i>'p'</i>	<i>p</i>	
<i>'probably p'</i>	$\emptyset$	<i>S</i> considers <i>p</i> likely enough to support its inclusion in the CG

As shown in (43), in a regular assertion (assuming no additional presuppositions), the uttered proposition constitutes the publicly advanced, asserted content. By contrast, in a proposal, no proposition is publicly advanced. Rather, the proposal is entirely speaker-oriented, expressing the speaker's support for a proposition, which remains merely publicly displayed.

The speaker's attitude of support or bias is not itself available for negation or rebuttal, suggesting that it patterns with presuppositional felicity conditions (see also e.g., Korotkova 2020 on evidentiality). Then by the Gricean Maxim of Quality, the speaker's overt support for a proposition gives rise to the implicature that the proposed proposition is true. This, in turn, makes the proposition itself available for rebuttal (see (42a) and §3.5). However, because no proposition is publicly advanced, there is no suitable target for full acceptance, accounting for the infelicity of responses such as (42b).

Following Romero and Han (2004) and Frana and Rawlins (2019)'s formal analysis of CG management, (44) presents the formal analysis of a proposal:

$$\begin{aligned}
 (44) \quad & \llbracket \text{probably } p \rrbracket^{c,w}, \text{ defined for } p, c, w \text{ only if } \underbrace{p \notin \text{CG}_w}_{p \text{ is not yet included in the CG}} \quad \wedge \\
 & \underbrace{\forall w' \in \text{Epi}_{\text{SPEAKER}_c}(w) : (}_{\text{The speaker is sure that}} \underbrace{\mu_{\text{PROB}_{w'}}(p \mid \text{CG}_w) \geq \theta_{c,w'}}_{p \text{ is sufficiently likely in the speaker's view}} \wedge \\
 & \underbrace{\exists w'' \in \text{Conv}_{\text{SPEAKER}_c}(w') : (p \in \text{CG}_{w''})}_{\text{in some worlds satisfying the conversation goals, } p \text{ is included in the CG}} \quad \bigg)
 \end{aligned}$$

In (44), CG_w is the common ground at a world w in a given context, i.e., the set of propositions that the participants in the conversation mutually assume to be true at w (Stalnaker 1978). $\text{Epi}_{\text{SPEAKER}_c}(w)$ is the set of the speaker's epistemic worlds, and $\text{Conv}_{\text{SPEAKER}_c}(w')$ is the set of worlds in which the speaker takes the conversational goals to be satisfied. The speaker publicly displays a proposition p such that (i) in the speaker's epistemic worlds, p is sufficiently likely given the current CG_w , and (ii) p is not excluded from the CG in every world satisfying the speaker's conversational goals. Equivalently, in some worlds satisfying the speaker's conversational goals, p is included in the CG.

The infelicity of semantically negative modal adverbs (see (45)) follows from their failure to satisfy the restriction that the proposed proposition p be sufficiently likely in the speaker's epistemic worlds. A speaker should not support

a proposition that she considers unlikely. The infelicity of *doubtedly* (see (45b)) further confirms that this restriction is semantic rather than morphological.

- (45) a. #Improbably it rained.
b. #Doubtedly it rained. (see also Bellert 1977)

Under the current analysis, the status of a proposal is presupposition-like: the proposed proposition is embedded within an attitude report, and this attitude report cannot be negated and is therefore treated as presuppositional. This explains why sentences containing modal adverbs resist negation and questioning, thereby demonstrating PPI-like behavior.

The current analysis also accounts for the infelicitous example discussed by Nilsen (2004) (see (33b), repeated here as (46)), which contrasts with the felicitous example in (47). As sketched in (47), an attitude holder may regard both p and $\neg p$ as sufficiently likely, and this gives rise to no contradiction. (47) simply means that Bill is uncertain whether he will win.

However, as illustrated in (46), although a speaker may regard both p and $\neg p$ as sufficiently likely, she cannot propose (i.e., publicly display and support) both p and $\neg p$. Doing so would commit the speaker to two incompatible conversational goals: supporting the consideration of p for inclusion in the CG, and supporting the consideration of $\neg p$ for inclusion in the CG. Therefore, under the current analysis, the infelicity of (46) arises not from the speaker's beliefs or probability estimates, but from the incompatibility of the speaker's conversational goals.

- (46) #Le Pen will possibly win, even though he certainly won't.
 $\exists w'' \in \text{Conv}_{\text{SPK}_c}(w') : p \in \text{CG}_{w''} \wedge \exists w''' \in \text{Conv}_{\text{SPK}_c}(w') : \neg p \in \text{CG}_{w'''}$
 (47) Bill thinks it's possible that he'll win, but it's also possible that he won't.
 $\forall w' \in \text{Epi}_{B_c}(w) : (\mu_{\text{PROB}_{w'}}(p \mid \text{CG}_w) \geq \theta_{c,w'} \wedge \mu_{\text{PROB}_{w'}}(\neg p \mid \text{CG}_w) \geq \theta_{c,w'})$

To sum up, the current analysis treats modal adverbs as signaling a speech act of proposal, rather than modifying a proposition or an assertion. This analysis accounts for the speaker-orientedness of modal adverbs, their PPI-like behavior, and the constraints governing proposals and responses to them.

3.3 Rejection as a speech act

The speech-act-based analysis of modal adverbs developed above can be naturally extended to account for the behavior of corrective constructions, in particular their requirement for sentential negation. Specifically, corrective constructions signal another speech act, rejection, as characterized in (48):

- (48) **The semantics of a rejection, *bìng bù* p , or *not* p (*but* q):**

	Publicly advanced (asserted)	Publicly displayed (presupposed)
regular ' <i>not</i> p '	$\neg p$	
' <i>bìng bù</i> p '	$\neg p$	despite considering p likely,
' <i>not</i> p (<i>but</i> q)'	$\neg p(\wedge q)$	S excludes it from the CG

As shown in (48), both regular negation and corrective constructions express a publicly advanced, asserted proposition: the negation of p .

However, corrective constructions additionally involve a speaker-oriented component: despite considering p likely, the speaker insists on rejecting it from the CG. In other words, rejection is a speech act that presupposes potential disagreement among the participants. When no salient disagreement exists, the speech act of rejection lacks a motivation and becomes infelicitous (see the comparison between regular negation and rejection in (49) and (50), especially (50b)).

- (49) A: Did it rain? / It rained. (non-biased yes/no question or assertion)
 a. B: It didn't rain. ✓Regular negation
 b. B: It didn't rain, but snowed. ✓Rejection
- (50) A: It didn't rain, did it? Bias $\rightsquigarrow$ there is no disagreement
 a. B: It didn't rain. ✓Regular negation
 b. #B: It didn't rain, but snowed. #Rejection

Thus, (51) presents the formal analysis of the two corrective constructions:

$$(51) \quad \left\{ \begin{array}{l} \llbracket \text{not } p \text{ but } q \rrbracket^{c,w} \stackrel{\text{def}}{=} \neg p \wedge q \\ \llbracket \text{b\`ing b\`u } p \rrbracket^{c,w} \stackrel{\text{def}}{=} \neg p \end{array} \right\}, \quad \text{defined for } p, c, w \text{ only if}$$

$$\underbrace{\forall w' \in \text{Epi}_{\text{SPEAKER}_c}(w) : \left(\underbrace{\mu_{\text{PROB}_{w'}}(p \mid \text{CG}_w) \geq \theta_{c,w'}}_{\substack{\text{The speaker is sure that } p \text{ is sufficiently likely given the CG} \\ \forall w'' \in \text{Conv}_{\text{SPEAKER}_c}(w') : (p \notin \text{CG}_{w''})}} \right)}_{\text{in all worlds satisfying their conversation goals, } p \text{ is not CG}}$$

In (51), the speaker rejects a proposition p such that (i) in the speaker's epistemic worlds, p is sufficiently likely given the current CG_w , but (ii) p is excluded from the CG in every world satisfying the speaker's conversational goals. This universal modality makes the negation of p publicly advanced as asserted content, rather than merely publicly displayed.

As CG-management operators, *b\`ing b\`u* p and *not* p *but* q operate on the entire proposition p , blocking its inclusion in the CG. Thus, sentential negation in a rejection does not merely reverse the truth value of p ; rather, the act of rejection changes p 's discourse status by preventing it from becoming publicly accepted. Because rejection targets the discourse status of a proposition rather than its truth conditions, sentential negation cannot combine with proposition-internal negation to yield an equivalent positive sentence. For the same reason, the sentential negation in corrective constructions cannot be replaced by morphologically encoded proposition-internal negation.

Under the current analysis, the speaker's assessment that p is highly likely coexists with the speaker's rejection of the inclusion of p in the CG. Thus, a corrective construction is used precisely when the speaker acknowledges that the inclusion of p in the CG is an expected conversational development but signals disagreement. The speech act of rejection thereby prevents a previously expected direction of the conversation from being established.

To sum up, the analysis of corrective constructions follows naturally from the speech-act perspective developed for modal adverbs. While a proposal publicly displays and supports a proposition, making it publicly visible and available for consideration for inclusion in the CG (cf. Farkas and Bruce 2010’s ‘Table’), a corrective construction performs the opposite conversational operation: it rejects the inclusion of a proposition that is already publicly visible, whether it is a proposed proposition, an asserted but not yet CG-included proposition, or simply an implicature. In some cases, the use of a corrective construction requires interlocutors to accommodate such public visibility. The speech-act perspective thus provides a unified account of the opposite polarity restrictions observed for modal adverbs and corrective constructions.²

3.4 Assertion vs. proposal vs. rejection

§3.2 and §3.3 developed analyses of proposals and rejections by comparing them with ordinary assertions. This subsection highlights the key properties that distinguish these three speech acts and clarifies their relationships.

(52) **The comparison of three speech acts:**

publicly speaker- advanced oriented		strength of the attitude	
I assert $p/\neg p$	$p/\neg p$	no	—
I propose p	$\emptyset$	yes	weak (existential modality)
I reject p	$\neg p$	yes	strong (universal modality)

First, both proposals and rejections contain a speaker-oriented component that resists negation and questioning, and therefore behaves like non-at-issue meaning (see (52)). By contrast, assertions lack such a speaker-oriented component. In this respect, proposals and rejections resemble evidential phenomena: they encode the speaker’s attitude while simultaneously managing the discourse status of a proposition. Together, these three speech acts reflect different stages of a proposition, from a speaker’s private belief to public visibility, readiness for interlocutor response, and ultimately inclusion in or exclusion from the CG.

Second, both proposals and rejections involve propositions that the speaker regards as sufficiently likely. In proposals, this likelihood reflects the speaker’s own epistemic assessment and motivates the proposal (i.e., public display) of the proposition. In rejections, by contrast, it reflects the speaker’s expectation that other interlocutors are inclined to accept the proposition, thereby motivating the counterexpectational speech act of rejection.

² An additional remark is that the polarity restrictions discussed here are not entirely homogeneous. The scope behavior of modal adverbs, their incompatibility with yes/no questions, and the requirement for sentential negation in corrective constructions all follow from the speech acts themselves, which operate on whole propositions. By contrast, the absence of forms such as *improbably* and *doubtedly* follows from the semantic licensing condition that a proposal requires the speaker to regard the proposition as sufficiently likely.

Finally, proposals and rejections differ in the strength of attitude. Proposals involve existential modality: the speaker supports, but does not require, the proposition's inclusion in the CG. Rejections, by contrast, involve universal modality: the speaker requires the proposition's exclusion from the CG. Consequently, proposals merely make a proposition publicly visible, whereas rejections publicly advance its negation as asserted content.

3.5 The interaction between proposals and rejections

One consequence of the current analysis is that rejection can target any publicly visible proposition, including one that is only implicated rather than asserted. Since proposals make propositions publicly visible, proposed propositions can likewise be rejected.

As shown in (53a), although A's proposal does not publicly advance an asserted proposition, the speaker's support for p gives rise, via the Maxim of Quality, to the implicature that p is true. This implicature satisfies the licensing condition for rejection, allowing B naturally to reject the implied proposition.

By contrast, the response in (53b), which embeds a proposal within a corrective construction, is infelicitous (setting aside metalinguistic negation). Under the current analysis, *probably* mainly signals the speech act of proposal rather than contributing a proposition. Accordingly, rejection cannot target the speech act of proposal. The infelicity of (53b) also suggests that the speaker-oriented part contributed by a proposal is itself unavailable as the target of rejection.

- (53) A: It will probably rain.
 a. B: It's not that it will rain, but that it will snow.
 b. #B: It's not that it will probably rain, but that it will rain.

The converse question is whether a rejection can itself be embedded within a proposal. In principle, rejection contributes the asserted proposition $\neg p$, which should be available for further speech acts. Nevertheless, the English example in (54b) and the Chinese example in (55) are both noticeably degraded compared with their counterparts without a modal adverb. One possible explanation is that the strong disagreement conveyed by a rejection makes it difficult for the speech act of proposal to weaken the negated proposition. In other words, the existential modality of proposal conflicts with the universal modality of rejection.

- (54) A: It will rain.
 a. B: It's not that it will rain, but that it will snow.
 b. ?B: Probably it's not that it will rain, but that it will snow.

- (55) Context: Mary seems to think that it will rain. So I tell her:

? míngmíng bìng bú huì xià yǔ
 evidently BING NEG will fall rain

≈ 'Evidently, it will not rain.'

Overall, a proposition that is merely proposed can be naturally rejected. However, embedding a modal adverb within a corrective construction, or vice versa, is degraded. This degradedness provides further evidence that these constructions contribute more than propositional content: they encode speech acts whose speaker-oriented components resist embedding.

4 Discussion

In formal semantics, assertions have been extensively studied since [Stalnaker \(1978\)](#). Asserted content contributes to updating the CG. [Farkas and Bruce \(2010\)](#) further emphasize the proposal-like nature of assertion: what is asserted does not update the CG automatically, and the update of the CG involves additional conversational moves, e.g., confirmation, rejection.

The current work inherits the view of [Farkas and Bruce \(2010\)](#) and makes a more fine-grained distinction between (i) asserted content that is publicly advanced, ready for interlocutor response, and (ii) proposed content that is merely publicly visible. This distinction is motivated by linguistic expressions such as modal adverbs, which require teasing apart (i) the speaker’s epistemic attitude toward a proposition (e.g., considering it highly likely or worth mentioning) and (ii) the speaker’s expectation about how interlocutors will respond to a proposition and whether it will ultimately be included in or excluded from the CG.

This perspective has broader implications for the analysis of discourse-related meaning. First, it suggests a closer connection between modal adverbs and evidentiality. Evidentials are expressions that address the source of information, e.g., direct perception, hearsay. Like evidentials, modal adverbs encode speaker-oriented information that (i) motivates an utterance and (ii) influences the status of a proposition in discourse. The fine-grained distinction adopted in the current work can provide a new perspective on whether and how evidential markers may signal different speech acts.

Second, the current analysis invites a reexamination of the relationship among declaratives, interrogatives, and imperatives. As illustrated at the beginning of the paper (see (4)), a proposal such as *It will probably rain* can be paraphrased by an imperative such as *Let’s consider that it will rain*, whereas a rejection can be paraphrased by an imperative such as *Don’t believe that it will rain*. More generally, proposals can be regarded as occupying an intermediate position between assertions and yes/no questions: they make propositions publicly visible without publicly advancing them as assertions do. This perspective may help clarify the shared and distinct discourse functions of different sentence types and shed light on the underlying mechanisms of human cognition.

Moreover, natural language contains a rich inventory of speaker-oriented sentential adverbs whose discourse functions remain poorly understood (e.g., *honestly*, *impossibly*; see [Bellert 1977](#)). The current work raises the possibility that different classes of such expressions correspond to distinct speech acts, opening a broader research agenda on the relationship between speaker-oriented meaning and discourse management.

5 Conclusion

This paper investigates modal adverbs and corrective constructions, arguing that they are speaker-oriented, CG-management operators, signaling speech acts of proposals and rejections. This speech-act perspective provides a unified account of their polarity restrictions and highlights a finer-grained distinction between publicly visible and publicly advanced propositions. More broadly, the analysis suggests a richer inventory of discourse-management speech acts beyond ordinary assertion.

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